



IMPORTANT

- **Test Type**
Game Programming Test
- **Task Duration**
Within 12 hours
- **Maximum Submission Time Allowed**
Up to 24 Hours
- **If fail to submit the test within the Stipulated Time Frame**
The Application will be Disqualified and considered Rejected

Task 1: AR Detection & Car Placement (Using AR Core or Vuforia)

- Create an AR application where a **car model** is placed in the real world using AR detection and can be interacted with using on-screen controls.
- Car Asset
(<https://drive.google.com/drive/folders/1k-YbqX1SssERF2IAZnezRImPRpeJnxY2?usp=sharing>)

TASK DESCRIPTION

1. The car model should be placed accurately and remain stable after detection.
2. Create on-screen UI buttons for triggering car interactions.
3. Implement functionality to start and stop tyre rotation animation.
4. Implement open and close doors, hood, boot animation through UI controls.
5. All animations should play smoothly and proper state handling should be maintained
6. Scripts should be organized properly with separate handling for AR logic, animation logic, and UI interaction.

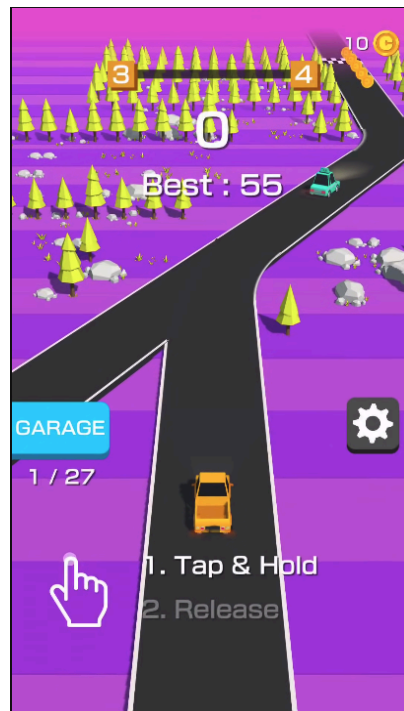
GAME CONTROL

1. Rotate Tyre
2. Stop Tyre Rotation
3. Open/Close Door
4. Open/Close Hood/Boot

Task 2 : Traffic Run Demo

VISUAL AND GAME REFERENCE:

- [Video](#)



- Create a prototype of the mechanism shown in the video. Try to replicate it as closely as possible. You can use placeholder(dummy) graphics/objects for this.
- You can use different 3D Items (<https://www.kenney.nl/assets/car-kit>) or Unity cubes as gameplay objects. (Free to use any 3D assets)

TASK DESCRIPTION

1. Gameplay logic
 - a. Create player vehicle movement on predefined road paths.
 - b. Implement a waypoint or spline-based path following system.
 - c. Create AI traffic vehicles moving continuously on assigned routes.
 - d. Add looping traffic behavior for continuous gameplay.
2. Game fail/win logic with placeholder UI
3. Add vehicle crash handling and level restart functionality.
4. Score logic

GAME CONTROL

1. Same as the video
 - a. Implement **Tap/Hold to Move** vehicle control system.

- b. Vehicle should **start moving on user input** and **stop when input is released**
- c. Smooth **acceleration and deceleration** for realistic movement

LEVEL DETAILS (Total 3 levels)

1. Generate 3 levels with an easy, medium and hard difficulty

FINAL SUBMISSION INSTRUCTIONS

1. At the end of the test, your code, logic, work and approach will be evaluated. Your creativity can also make a huge impact, we are open if you put some creative ideas other than this.
 - a. **Submission Format:** Source File
 - b. **File Name Should be:** YOUR NAME_DDMMYY (i.e. Rahul_201224)
 - c. **Min Unity Version should be** v2022.3.62f2, Unity 6 or later

Important Note:

- Both tasks are mandatory and compulsory to complete. The project will consist of a single APK, but it must contain two separate scenes for both tasks. Each task should function independently in its respective scene, and completion of both tasks is required as part of the final deliverable.
- You may use utility scripts such as **Tween** or **Singleton**. However, the use of ready-made Unity packages, AI tools (including ChatGPT or similar), or pre-written scripts to implement the required functionality is strictly prohibited. Any violation of this rule will result in **immediate disqualification**.